WORKED SOLUTION

# STEP II 2010, Question 5

Pure • Vector Angles and Cones

**STEP II 2010, Question 5**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q5

## The key idea

Everything here comes down to one fact: the angle $\theta$ between a line with direction $\mathbf{d}$ and a vector $\mathbf{a}$ satisfies $\cos\theta = \dfrac{\mathbf{d}\cdot\mathbf{a}}{|\mathbf{d}|\,|\mathbf{a}|}$. "Inclined equally" and "inclined at angle $\alpha$" turn straight into equations in the components of $\mathbf{d}$. For the final part, notice that the Cartesian equation has the same shape as the relationship you have just found. A set of lines through $O$, all making the same angle with a fixed line, is a cone.

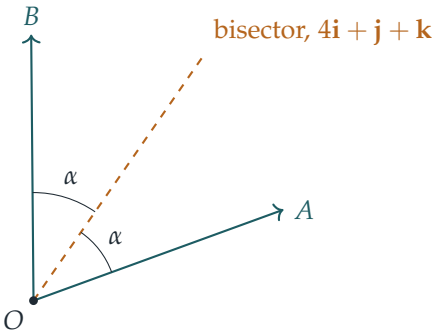
## Solution

Write $\mathbf{a} = \mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i}-\mathbf{j}-\mathbf{k}$, so that $|\mathbf{a}| = \sqrt{3}$ and $|\mathbf{b}| = \sqrt{27} = 3\sqrt{3}$. Also $\mathbf{a}\cdot\mathbf{b} = 5-1-1 = 3$. Hence

$$\cos 2\alpha = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}|\,|\mathbf{b}|} = \frac{3}{\sqrt{3}\cdot 3\sqrt{3}} = \frac{1}{3}.$$

Since $\cos 2\alpha > 0$, the angle $2\alpha$ is acute, so $\alpha$ lies between $0$ and $\frac{\pi}{4}$. We will need

$$\cos^2\alpha = \tfrac{1}{2}(1+\cos 2\alpha) = \tfrac{2}{3}.$$




(The picture shows the plane $OAB$ only schematically.)

**Part (i)**

Let $\mathbf{d} = m\mathbf{i} + n\mathbf{j} + p\mathbf{k}$ be the direction of $L_1$. The cosines of the angles that $\mathbf{d}$ makes with $OA$ and $OB$ are

$$\frac{\mathbf{d}\cdot\mathbf{a}}{|\mathbf{d}|\sqrt{3}} = \frac{m+n+p}{\sqrt{3}\,|\mathbf{d}|} \qquad \text{and} \qquad \frac{\mathbf{d}\cdot\mathbf{b}}{|\mathbf{d}|\,3\sqrt{3}} = \frac{5m-n-p}{3\sqrt{3}\,|\mathbf{d}|}.$$

Equal angles means equal cosines, since cos is one-to-one on $[0, \pi]$. Multiplying through by $3\sqrt{3}\,|\mathbf{d}|$ gives

$$3(m+n+p) = 5m - n - p \quad \Longleftrightarrow \quad m = 2(n+p).$$

**Ask yourself** A line has no preferred direction. What if the line makes angle $\theta$ with $OA$ but $\pi - \theta$ with $OB$?

If we take "the angle between two lines" to mean the acute one, then the cosines need only be equal up to sign. The other sign gives $3(m+n+p) = -(5m-n-p)$, that is, $4m + n + p = 0$. These lines lie in the plane through $O$ perpendicular to the bisector, which contains the external bisector of $\angle AOB$. The intended answer is $m = 2(n+p)$, but it's worth knowing that the second family exists.

For the bisector of $\angle AOB$, we add the *unit* vectors along $OA$ and $OB$. The sum of two unit vectors bisects the angle between them because they span a rhombus. So

$$\frac{\mathbf{a}}{\sqrt{3}} + \frac{\mathbf{b}}{3\sqrt{3}} = \frac{3\mathbf{a}+\mathbf{b}}{3\sqrt{3}} = \frac{8\mathbf{i}+2\mathbf{j}+2\mathbf{k}}{3\sqrt{3}},$$

and we can take $m = 4$, $n = 1$, $p = 1$ (or any non-zero multiple). As a check, $4 = 2(1+1)$, so this direction satisfies the relation above.

**Part (ii)**

Let $\mathbf{e} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}$ be the direction of $L_2$. The line makes angle $\alpha$ with $OA$ when

$$\frac{u+v+w}{\sqrt{3}\,\sqrt{u^2+v^2+w^2}} = \pm\cos\alpha.$$

The $\pm$ allows for either sense along the line. Squaring removes the sign, and using $\cos^2\alpha = \frac{2}{3}$ we get

$$(u+v+w)^2 = 2(u^2+v^2+w^2).$$

Expanding the left-hand side as $u^2+v^2+w^2+2(vw+wu+uv)$ and rearranging gives the relationship

$$u^2+v^2+w^2 = 2(vw+wu+uv).$$

**Ask yourself** How does a condition on a *direction* turn into a *surface*?

Take any point $(x, y, z) \neq (0, 0, 0)$. It lies on exactly one line through $O$, namely the line with direction $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. By part (ii), that point satisfies

$$x^2 + y^2 + z^2 = 2(yz + zx + xy)$$

exactly when its line through $O$ makes angle $\alpha$ with $OA$. The origin also satisfies the equation. So the surface is the union of all lines through $O$ inclined at angle $\alpha$ to the line $OA$.

This is a **(double) right circular cone** with vertex at the origin and axis the line $x = y = z$ (the line $OA$). Its semi-vertical angle $\alpha$ satisfies

$$\cos\alpha = \sqrt{\tfrac{2}{3}}, \qquad \tan\alpha = \tfrac{1}{\sqrt{2}}, \qquad \alpha \approx 35.3^\circ,$$

which is half of $\angle AOB$. It has two nappes because $\mu$ runs over all real numbers. The bisector from part (i) lies on the cone: for $(4, 1, 1)$ we get $16 + 1 + 1 = 18 = 2(1 + 4 + 4)$.

## Checks

- The point $(1, 1, 0)$ satisfies $2 = 2(0 + 0 + 1)$. Its angle with $\mathbf{i} + \mathbf{j} + \mathbf{k}$ has cosine $2/(\sqrt{2}\sqrt{3}) = \sqrt{2/3}$, as it should.
- Numerically, $\arccos(1/3)/2 = 0.61548$ rad, and the bisector $4\mathbf{i} + \mathbf{j} + \mathbf{k}$ makes angle $0.61548$ rad with both $OA$ and $OB$.
- The identity $x^2 + y^2 + z^2 - 2(yz + zx + xy) = (x^2 + y^2 + z^2)(2 - 3c^2)$ holds, where $c$ is the cosine of the angle with $\mathbf{i} + \mathbf{j} + \mathbf{k}$. So the left-hand side vanishes exactly when $c^2 = \frac{2}{3}$.

## Common slips

- Bisecting with $\mathbf{a} + \mathbf{b}$ instead of the sum of *unit* vectors. Since $|\mathbf{b}| \neq |\mathbf{a}|$, $\mathbf{a} + \mathbf{b} = 6\mathbf{i}$ is not the bisector.
- Using $\cos\alpha = \frac{1}{2}\cos 2\alpha$ or similar. You need the double-angle formula, $\cos^2\alpha = \frac{1}{2}(1 + \cos 2\alpha)$.
- Describing the surface just as "a cone" without giving its vertex, axis and semi-vertical angle.

## Beyond STEP

The equation of the cone is a *quadratic form*, $\mathbf{x}^{\mathsf{T}} M\mathbf{x} = 0$ with $M$ a symmetric matrix. In first-year linear algebra you classify such surfaces by the eigenvalues of $M$. Here $M$ has diagonal entries 1 and off-diagonal entries $-1$, so its eigenvalues are $-1$ (along $\mathbf{i} + \mathbf{j} + \mathbf{k}$) and 2 (twice). One negative and two equal positive eigenvalues are exactly the signature of a circular cone about the eigenvector of the odd one out. Strang's

*Introduction to Linear Algebra* develops this in its treatment of symmetric matrices and quadratic forms.

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